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Enhancing CO₂ Removal and Mitigating Methane Emissions Through Higher-Selectivity Membranes

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Abstract

Nestled within Field R, Platform X serves as a cornerstone in the critical task of CO₂ removal from high-CO₂ feed gas to meet stringent sales gas specifications. The two-stage membrane system effectively separates CO₂ but results in continuous venting of CO₂-rich streams containing methane, contributing to greenhouse gas emissions.

This paper showcases the successful implementation of higher selectivity membranes at Platform X to reduce methane in the vent stream. Desktop simulations assessed the impact of high-selectivity membranes in both the Primary and Secondary Membrane systems. This was followed by a two-phase field trial: Phase 1 tested high-selectivity membranes in a Primary Membrane skid pre-permeate stream typically routed to vent, while Phase 2 upgraded the Secondary Membrane system.

Post-changeout sampling showed a 3% increase in vent CO₂ purity to 96%, hydrocarbon recovery improved by 3% to 97.5%, and ~4 MMscfd of hydrocarbon was reclaimed equating to 840 ktCO₂e/year of GHG reduction. The payback period was approximately 12 months.

A key insight is that enhanced vent CO₂ purity supports future CCS readiness, positioning Platform X for sustainable carbon management aligned with industry decarbonization goals.

Process Overview and Baseline System

With reference to [Figure 1](#), at the core of Platform X lies a two-stage membrane separation system, comprising a Primary Membrane Skid, Permeate Compressors, and a Secondary Membrane Skid. Both the Primary and Secondary Membrane Skids utilize prefabricated membrane tubes that house spiral-wound membrane elements, which are designed to maximize surface area and promote uniform gas flow distribution. These membranes operate based on the differential pressure ([Lokhandwala et al, 2010](#)) between the high-pressure feed side and the low-pressure permeate side to selectively transport CO₂ across the membrane surface. Less permeable components such as methane and heavier hydrocarbons remain on the high-pressure residue side ([Beronich, 2006](#)).

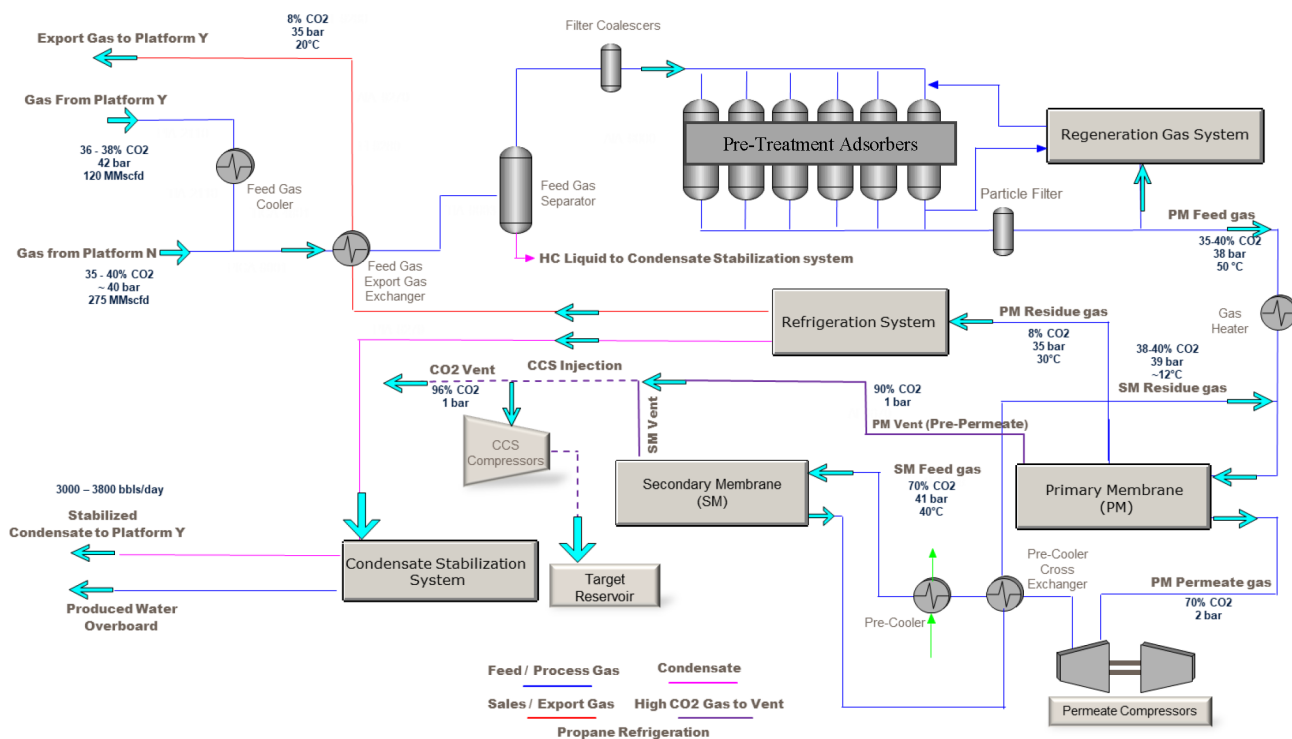


Figure 1—Platform X Simplified Process Flow Diagram

This two-stage arrangement is purposefully designed to enhance separation efficiency and hydrocarbon retention. The primary stage is responsible for bulk CO₂ removal forming the first barrier to meet the sales gas specification. The secondary stage, on the other hand, is primarily designed for hydrocarbon recovery. Its residue stream, which is rich in methane, is recycled back to the primary membrane feed, thereby improving overall hydrocarbon retention while still contributing to CO₂ removal efficiency.

The pretreatment section plays a critical role in ensuring membrane performance and longevity (Abdessemed et al., 2000). It conditions the feed gas by removing water vapor, mercury, and heavy hydrocarbons using a temperature swing-based adsorption system with adsorbent vessels operating in rotating cycles. Post pretreatment section, the gas is cooled via cross-exchangers and passed through high-efficiency filter coalescers and particle filters to eliminate entrained liquids and aerosols. This ensures that the feed entering the membrane elements is dry and free of contaminants that could damage or foul the membranes. Following pretreatment, the gas is heated in membrane preheaters and mixed with the Secondary Membrane residue stream before entering the Primary Membrane section. CO₂ permeates through the membrane to the low-pressure side, while methane and heavier hydrocarbons remain in the high-pressure residue stream. The Primary Membrane handles bulk CO₂ removal, with its permeate stream collected at two points near the feed and the residue ends to enable optimization of hydrocarbon retention. Portions of this permeate are either vented or recompressed depending on system loading and compression capacity.

The permeate stream from the Primary Membrane, which contains approximately 70 mol% CO₂, is compressed and routed to the Secondary Membrane for further separation. In the Secondary Membrane, additional hydrocarbons are recovered into the residue stream, which contains around 40 mol% CO₂ and is recycled back to the Primary Membrane feed. The CO₂-rich permeate from the Secondary Membrane, now at higher purity, is continuously vented to atmosphere at approximately 100 kPag, ensuring stable dispersion conditions and minimizing methane loss.

Overall, Platform X processes approximately 400 MMscfd of feed gas at ~40 bar with 35-38 mol-% CO₂. Prior to membrane enhancement, the system vented about 150 MMscfd of permeate gas containing 93 mol%

CO₂ and trace hydrocarbons. Membrane performance is defined by two interdependent parameters: capacity, which determines how effectively CO₂ can be removed to meet product specifications, and selectivity, which dictates the membrane's ability to preferentially allow CO₂ to permeate while rejecting hydrocarbons. Selectivity, in particular, governs the degree of methane migration into the vent stream and has a direct impact on both hydrocarbon recovery and GHG emissions (Fan, 2024).

Simulation Methodology

A structured simulation and evaluation approach was adopted to determine the optimal deployment of highly CO₂-selective membranes within Platform X's two-stage membrane system. The methodology involved two key stages:

1. Identifying whether the Primary Membrane (PM) or Secondary Membrane (SM) would offer greater improvement in hydrocarbon recovery and emission reduction upon installation of advanced membranes
2. Determining the most effective membrane configuration to maximize system performance at the selected stage.

The performance evaluation was conducted through a series of desktop simulations using a proprietary process simulator provided by the membrane technology vendor. Due to commercial sensitivity, the specific name and version of the software are not disclosed. The simulator incorporates vendor-specific membrane performance correlations derived from experimental data and was used to model multi-stage CO₂ separation across varying feed conditions and membrane configurations. The membrane behavior was modeled based on selective permeation principles (Khoros & Zhang, 2017), incorporating key parameters such as feed gas rate and composition, membrane surface area, feed and permeate pressures, and gas temperatures.

For consistency across all simulation cases, several baseline assumptions were applied. The feed gas flowrate was set at approximately 400 MMscfd at a pressure of around 40 bar, with a CO₂ concentration ranging between 35 and 38 mol%. The Primary Membrane permeate was modeled with a CO₂ concentration of approximately 70 mol%, while the Secondary Membrane residue stream, after recycling, contained about 40 mol% CO₂. Venting from the Secondary Membrane permeate was maintained at approximately 100 kPag to simulate actual operating practice. Unless otherwise specified, a constant membrane surface area was assumed throughout the simulations to ensure baseline comparability. Additionally, no significant modifications to the upstream pretreatment facilities or downstream compression systems were considered during the simulation scope, and all cases were evaluated under steady-state operating conditions.

Initial simulation runs explored the application of higher selectivity membranes at both the Primary Membrane (PM) and Secondary Membrane (SM) stages. For the PM evaluation, sensitivity cases were performed by incrementally replacing standard elements with higher-selectivity membranes at various permeate collection points. These simulations highlighted that while CO₂ purity improved in the permeate stream, the resulting increase in residue CO₂ concentration adversely impacted the system's ability to meet the 8 mol% sales gas specification. Furthermore, any significant rise in recycle CO₂ load to the PM inlet limited overall removal capacity, requiring tighter control of pressure and flow balancing. Though technically feasible, this path presented operational complexity and a narrower optimization margin.

In contrast, simulations for the SM stage demonstrated stronger gains in both vent stream CO₂ purity and hydrocarbon recovery. Because the SM permeate is directly vented, improving its CO₂ selectivity had an immediate and measurable impact on emissions reduction. Additionally, modifying the SM required less disruption to upstream configurations and allowed for implementation in modular phases. From an economic perspective, upgrading the SM also proved to be a more viable option due to its higher return on methane recovery and lower integration cost.

Following comparative evaluation, the SM stage was ultimately selected for selective membrane enhancement due to its direct influence on vent purity, higher operational feasibility, and favorable economic return. Subsequent sensitivity simulations focused on varying the number of high-selectivity elements installed at the SM stage, while holding membrane area, system layout, and operating conditions constant. Key performance metrics were tracked at each increment, including vent CO₂ purity, hydrocarbon recovery rate, methane loss to the atmosphere, and pressure drop across the membrane bank.

However, the simulations also revealed diminishing returns in performance improvement beyond a certain selectivity threshold. While early improvements showed substantial gains in methane recovery and vent purity, further increases in selectivity yielded marginal performance enhancement relative to the economic and operational complexity associated with higher-performance membrane materials. This inflection point was used to define the optimal selectivity range that maximized benefit while preserving operational feasibility and favorable economics.

The financial implications of each case were evaluated using the value of reclaimed hydrocarbons and estimated reductions in greenhouse gas emissions. Each scenario was associated with a projected payback period, calculated from the incremental revenue generated by hydrocarbon recovery. Based on these results, the optimal configuration was selected as the one that achieved a vent purity exceeding 96 mol% CO₂, hydrocarbon recovery above 97%, and a payback period of less than 12 months meeting both environmental and commercial objectives.

Implementation Strategy

The membrane enhancement initiative at Platform X was executed through a phased upgrade approach to minimize operational disruption while capturing key learnings along the way. Recognizing the challenges associated with offshore execution and the need to maintain production continuity, the project was structured in two stages. Phase 1 served as a pilot trial to assess performance response to highly CO₂-selective membranes at the Primary Membrane stage, while Phase 2 scaled the initiative to the Secondary Membrane, targeting system-wide improvements. This stepwise approach allowed for gradual implementation, validation of assumptions, and realtime tuning, supported by collaboration between offshore operations, engineering, and vendor teams. The campaign emphasized careful planning, safe execution, and close monitoring to ensure technical, environmental, and operational objectives were met.

Phase 1 Implementation at Primary Membrane

Phase 1 was executed by retrofitting a limited number of highly selective membrane elements into the pre-permeate section of a single Primary Membrane skid. This specific skid was selected based on its consistent routing of permeate to the vent system, allowing for a focused trial without impacting other parts of the plant. The trial offered valuable operational insight into the behavior of high-selectivity membranes under actual field conditions. Notably, the installation resulted in an increase in CO₂ concentration in the pre-permeate stream, while reducing its flowrate indicating more efficient separation and reduced methane slippage to the permeate side. While the scope was intentionally constrained to avoid broader system impacts, Phase 1 provided early confirmation of the technology's effectiveness and laid the groundwork for wider deployment.

Phase 2 Implementation at Secondary Membrane

Building on the findings from Phase 1, Phase 2 involved a full-scale membrane changeout campaign at the Secondary Membrane (SM) stage. This phase aimed to enhance vent CO₂ purity and improve overall hydrocarbon recovery at the facility. The changeout was executed sequentially, with each SM skid taken offline and upgraded individually to maintain operational stability. Prior to field execution, simulation results and vendor input guided the final membrane configuration. Execution involved detailed planning,

strict adherence to offshore safety procedures, and coordination across offshore and onshore teams. The installation process included the removal and replacement of membrane elements within pressure tubes, followed by leak testing and recommissioning. Post-changeout, each skid was closely monitored for integrity and performance. Field observations indicated a noticeable increase in CO₂ concentration in the SM permeate stream and a measurable reduction in overall vent flow validating the efficacy of the high-selectivity membranes and supporting their continued application across similar offshore gas systems.

Post-Changeout Monitoring and Dynamic Optimization

While the membrane changeout campaign represented a significant enhancement to the separation system, sustaining performance over time requires more than just equipment replacement. The operating environment at Platform X is inherently dynamic, with frequent variations in feed gas composition, flowrate, and export demands. These fluctuations directly influence membrane separation behavior. To ensure that hydrocarbon recovery remains consistently high and that CO₂ vent purity is maintained, a condition-based monitoring and optimization strategy was adopted following the membrane upgrade.

Post-changeout, the platform operations and onshore support teams implemented a close performance tracking program for each Secondary Membrane (SM) skid. This included real-time review of plant parameters and routine field sampling to validate gas composition, recovery trends, and vent stream characteristics. Rather than operating on fixed setpoints, the plant adopted a responsive control approach that allowed for fine-tuning based on prevailing process conditions.

Key parameters were actively managed to sustain optimal membrane performance. One area of focus was the dew point of the feed gas upstream of the Primary Membrane (PM). Maintaining a dry, stable feed through upstream separation and dehydration ensured membrane longevity and prevented hydrocarbon condensation within the elements. Plant operating pressure was also adjusted based on gas throughput and CO₂ concentration. Higher pressures typically enhanced CO₂ removal efficiency due to improved driving force, but excessive pressure could risk hydrocarbon slip. As such, pressure was optimized to strike a balance between CO₂ rejection and methane retention.

Temperature control played an equally critical role. Both PM and SM stages were monitored to ensure membrane temperatures remained within optimal permeation ranges. Too high a temperature can reduce selectivity by allowing methane to permeate more easily, while too low a temperature may impair permeability and risk condensation (Bakhtyari and Omidkhah, 2014). Adjustments were made based on daily thermal balance, heat exchanger performance, and ambient effects.

Finally, permeate backpressure was carefully managed at both the Primary Membrane (PM) and Secondary Membrane (SM) stages to maintain the desired transmembrane pressure differentials. In particular, a lower permeate pressure is intentionally maintained at the SM stage to promote stronger CO₂ driving force across the membrane, thereby enhancing separation and enabling the vent stream to achieve higher CO₂ purity. This strategy also facilitates continuous CO₂ removal without overloading the recompression system. By keeping permeate pressure at SM low, the plant ensures that CO₂ is preferentially transferred into the vent stream, effectively minimizing methane slip while supporting readiness for future carbon capture and storage (CCS) integration.

This continuous optimization process, grounded in both field observation and technical insight, has been essential to maintaining the gains achieved through the membrane upgrade. It reinforces that performance is not only determined by membrane design, but also by how the process is tuned and managed in real time. As feed conditions evolve, Platform X remains capable of dynamically adjusting its membrane operation to preserve high recovery, minimize emissions, and maintain reliability.

Results and Discussion

The membrane upgrade campaign at Platform X was executed in two distinct phases. In Phase 1, a limited number of high-selectivity membranes were installed at the Primary Membrane (PM) skid that was consistently routed to vent. This trial aimed to evaluate potential improvements at the earlier separation stage. Field results showed an increase in pre-permeate CO₂ concentration from 85% to 90%, accompanied by a 4% reduction in overall vent flow, confirming enhanced separation efficiency even with minimal membrane replacement. In Phase 2, the campaign shifted focus to the Secondary Membrane (SM) stage, where a larger-scale changeout was executed. This phase resulted in a rise in permeate CO₂ purity from 93% to 96% and a 17% reduction in total vent flow, significantly improving hydrocarbon recovery and positioning the system for CCS readiness. A summary of the results is presented in Table 1, with Figures 3–5 providing simplified visual representations.

Table 1—Platform X Membrane Changeout Phae 2 Campaign Results

Increasing Number Of Highly Selective To CO2 Membrane Elements					
SM Membrane Changeout	BASE	Skid X1 & Y1 (Trial Config)	Skid X2 (Optimized Config)	Skid Y2 (Optimized Config)	Skid X1 (Re-Optimized Config)
Changeout Timeline	Base	Dec 23	May 24	July 24	Oct 24
HC Recovered (MMscfd)		2.25 +2.25	3.00 +0.75	3.75 +0.75	4.5 +0.75
Vent Purity (CO ₂ %)	93	94.5	95	95.5	96
HCR (%)	95	96.4	96.7	97	97.5
GHG Reduction (ktco ₂ eq/annum)	Base	423	565	706	847
Cumulative Cost (Mil USD)	Base	2.2	3.04	4.63	5.67

Note 1: Flow measurements using the Annubar meter carry a typical uncertainty of ± 1.0 – 2.0% depending on calibration, installation conditions, and compensation factors. Online gas analyzer uncertainty is estimated at ± 1 mol% for CO₂ and CH₄.

Note 2: The methane abatement estimate of 4.5 MMscfd is based on actual field measurements conducted post-membrane upgrade, including manual gas sampling and validation using the platform's continuous flow monitoring system. However, due to dynamic variations in CO₂ content and feed gas flowrate, direct comparison of vent flow volumes before and after the upgrade is challenging. To provide a consistent basis for comparison, a constant vent flowrate of 150 MMscfd was assumed, while vent gas purity improved from 93 mol-% to 96 mol-%. The associated GHG reduction of 840 ktCO₂e/year was calculated using this representative purity improvement and a GWP of 28 (100-year horizon) for methane.

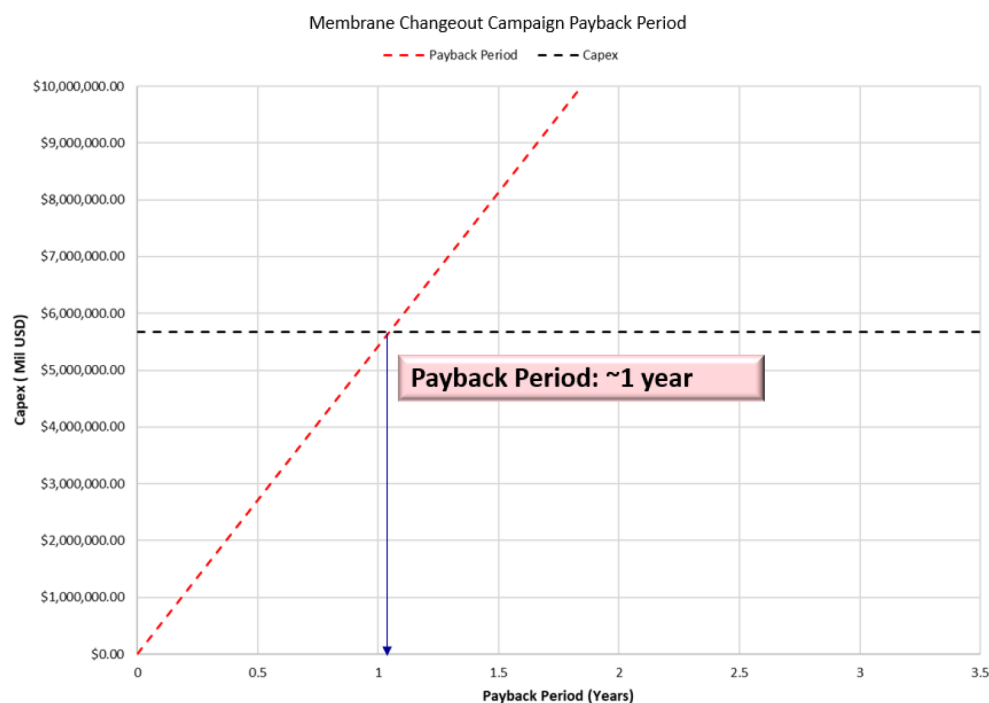


Figure 2—Platform X Membrane Changeout Campaign Payback Period Chart

Note: Hydrocarbon Sales Gas Price is reference to BRENT HSFO (Monthly Avg numbers) & Both Capex & Opex cost of Membranes have been taken into account in payback calculation.

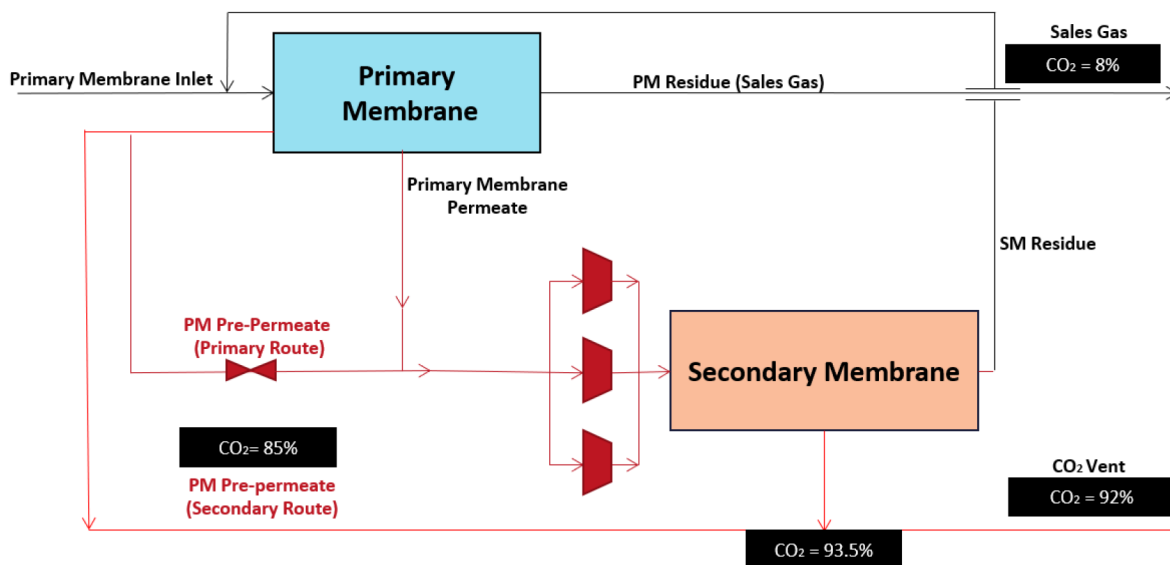


Figure 3—Base Case Scenario

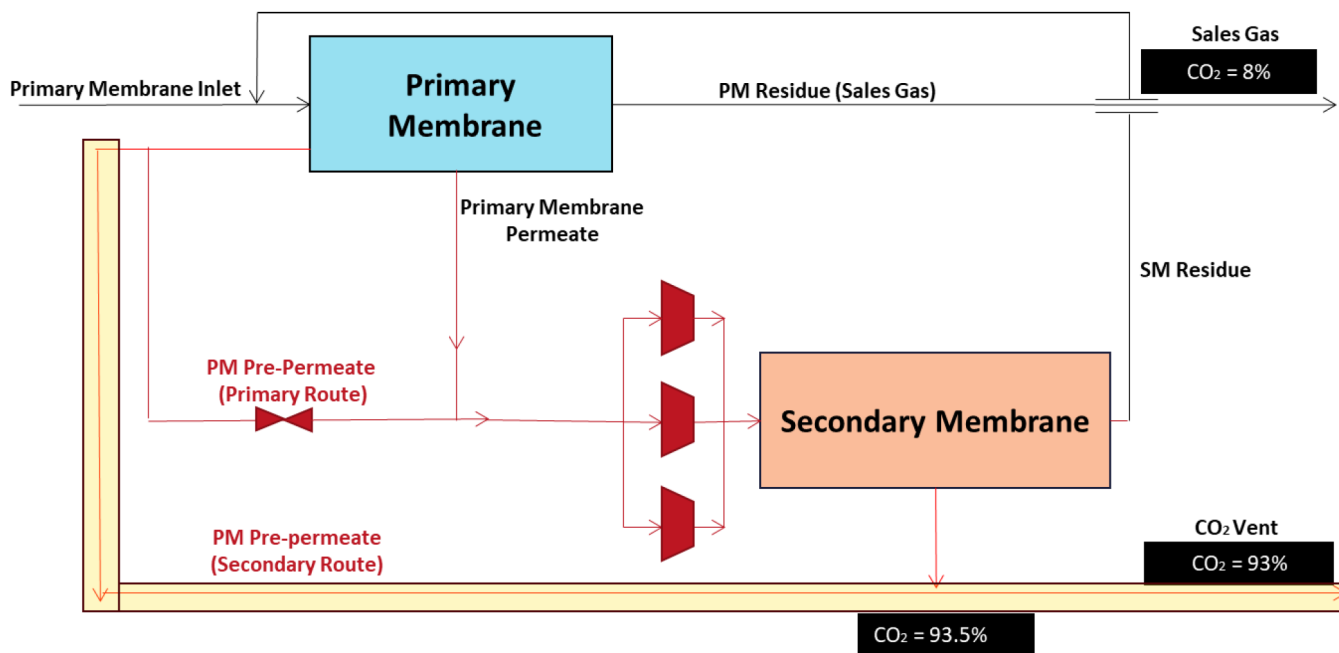


Figure 4—Phase 1 Trial Initiative on Primary Membrane

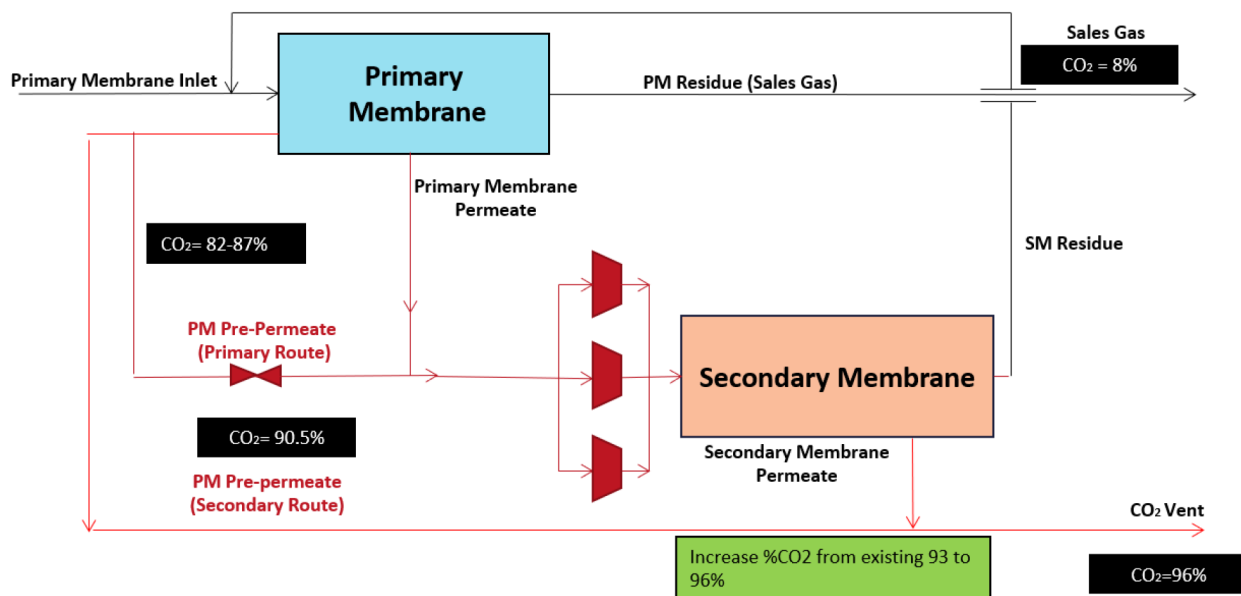


Figure 5—Phase 2 Trial Initiative on Secondary Membrane

An important insight from both simulation and field trials was the effect of inlet CO₂ concentration and CO₂ partial pressure on the performance of highly selective membranes. At the Secondary Membrane stage, where the feed gas CO₂ concentration is approximately 70 mol-%, increasing the number of high-selectivity membrane elements consistently improved permeate CO₂ purity. The elevated partial pressure of CO₂ at this stage enhanced the driving force for permeation, allowing the membranes to perform close to their ideal separation profiles. In contrast, at the Primary Membrane stage, where CO₂ concentration ranges between 35-40 mol-%, adding more selective membranes beyond a certain point reduced permeate CO₂ purity. This was due to insufficient driving force across the membrane, which compromised separation efficiency and increased methane slippage into the permeate stream. These findings highlight the importance of matching

membrane selectivity to feed gas composition underscoring that high-selectivity membranes yield greater performance when deployed downstream, where CO₂ content is already enriched.

CCS Readiness and Future Outlook

The successful enhancement of Platform X's membrane system has not only delivered tangible gains in hydrocarbon recovery and emissions reduction, but has also prepared the facility for integration into future carbon capture and storage (CCS) operations. A critical enabler of this readiness is the substantial increase in CO₂ purity in the vent stream, which now consistently achieves concentrations above 96 mol-% following the selective membrane upgrade and process tuning.

High-purity CO₂ is essential for reservoir injection (IEAGHG, 2011), as it improves compression efficiency and ensures compatibility with storage requirements. Through the deployment of highly selective membranes at the Secondary Membrane (SM) stage, Platform X has effectively transformed its vent stream into a high-concentration CO₂ stream suitable for direct injection into a subsurface reservoir.

Equally important is the platform's ability to minimize methane loss in the vent stream. Methane's significantly higher global warming potential compared to CO₂ (IPCC, 2021) makes it critical to demonstrate that upstream operations do not compromise the environmental integrity of CCS efforts. With a post-upgrade hydrocarbon recovery rate of 97.5%, Platform X has successfully reduced methane slip while simultaneously meeting CO₂ purity thresholds for geological storage.

The upgraded membrane system, coupled with active process control of key parameters such as pressure, temperature, and recycle flow, provides a stable operating platform for future CO₂ injection. This includes the potential for integration of a dedicated compression train and dehydration package to support continuous injection operations. The modular layout of the membrane units allows for targeted retrofits without significant production disruption.

Looking ahead, Platform X is well-positioned to contribute to national emissions reduction goals through the direct injection and permanent storage of CO₂. With high-purity CO₂ already being produced and vented, the facility now meets the critical upstream criteria to transition toward CCS operations, aligning with broader decarbonization strategies in the region.

Implementation Challenges and Operational Considerations

The implementation of high-selectivity membranes at Platform X presented several technical and operational challenges. One of the primary constraints was the trade-off between membrane selectivity and removal capacity. While higher CO₂/CH₄ selectivity effectively reduced methane slip and improved vent stream purity, it also resulted in elevated CO₂ concentration in the Secondary Membrane residue stream. This increased the CO₂ load recycled to the Primary Membrane, limiting the system's net removal capacity and making it more difficult to meet the 8 mol-% sales gas specification. To maintain removal efficiency under these conditions, operators were required to actively tune key process parameters particularly feed and permeate pressure, as well as membrane temperature. These adjustments directly influenced driving force and separation efficiency, helping to manage internal CO₂ buildup while preserving hydrocarbon recovery. The membrane replacement activity itself posed offshore execution challenges due to space constraints and physically demanding procedures, requiring phased implementation without disrupting production. Furthermore, the upgraded membranes had to integrate seamlessly with existing pretreatment and compression systems, offering little flexibility for hardware modification. Sustaining membrane performance post-changeout also required continuous monitoring and real-time operational responsiveness to account for fluctuations in feed gas flowrate, composition, and ambient conditions. These challenges underscored the importance of combining membrane design, execution strategy, and dynamic control to ensure stable, high-efficiency operation under real-world offshore conditions.

Conclusion

The membrane enhancement project at Platform X demonstrates how targeted process improvements can simultaneously advance environmental performance and economic return. Through a systematic evaluation and phased upgrade of the Secondary Membrane (SM) stage using highly CO₂-selective membranes, the platform achieved a 2.5% increase in hydrocarbon recovery and improved CO₂ purity in the vent stream from 93 mol-% to 96 mol-%. These gains translated into a methane recovery of 4 MMscfd and a greenhouse gas emissions reduction of approximately 800 ktCO₂e per year achieved with a payback period of under 12 months.

Beyond immediate performance improvements, the upgrade has positioned Platform X for long-term carbon capture and storage (CCS) readiness. With a high-purity CO₂ vent stream and minimal methane slip, the facility now meets key technical criteria for reservoir injection. The results of this project affirm the value of integrating advanced membrane technologies with dynamic operational strategies and offer a scalable model for emissions reduction and CCS preparedness across similar offshore gas processing assets.

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